

Session 2 · Deck 2.B3

Optimal Policy: Ramsey and Time Consistency

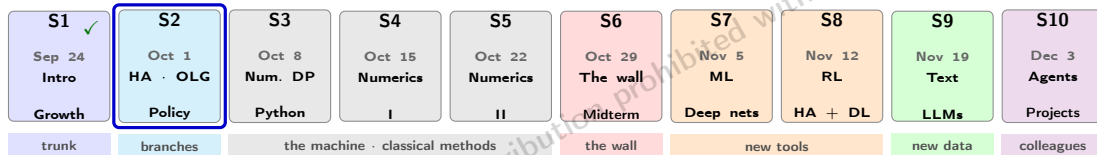
Zhigang Feng

Thursday, October 1, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

▼ you are here: 2.B3



Build it on paper → solve it on the machine → push it to the frontier with AI

Block B: 2.B1 aggregate risk and the distribution as a state · 2.B2 finite lives and cohorts · 2.B3 the government as a player: Ramsey and time consistency.

The Plan You Announce Is Not the Plan You Will Want to Keep

The capital levy

Date 0. The government announces low future capital taxes, so that households invest. They do.

Date 5. The capital is already in place. It cannot be un-built, and its short-run supply is essentially **zero**.

Taxing it now raises revenue and distorts *nothing* today. The date-5 government, re-optimizing, wants to tax it heavily — even though the date-0 government promised not to.

Time inconsistency

The optimal plan is not the plan that is optimal to *continue*. The conflict is not between good and bad governments — it is the **same** government, with the same objective, at two dates.

Kydland & Prescott (1977, *JPE*); Calvo (1978); Barro & Gordon (1983).

Households know this. If they expect the levy, they do not invest in the first place — so the promise is worthless precisely because it is not credible.

The Ramsey Problem: Optimal Taxes *with* Commitment

Suppose the government can bind itself at date 0 to an entire path $\{\tau_t^k, \tau_t^l\}_{t \geq 0}$. It maximizes household welfare subject to:

- the **resource constraint**;
- its own **budget constraint**;
- and — the distinctive part — the household's own **optimality conditions**.

The primal approach. Substitute taxes out using the household's first-order conditions and choose *allocations* directly, subject to one **implementability constraint**

$$\sum_t \beta^t [u_{c,t} c_t + u_{n,t} n_t] = u_{c,0} (1 + r_0) a_0,$$

which characterizes exactly the allocations some tax system can decentralize.

Why this is not an ordinary planner problem

The planner of 1.B1 chose an allocation subject to technology alone. The Ramsey government must also respect that **households optimize** — it is a Stackelberg leader, not a dictator.

Commitment is a strong assumption: the date-0 government binds every future government forever, with no enforcement mechanism in the model.

Commitment Adds a State Variable

Attach a multiplier Φ to the implementability constraint.

The household's intertemporal condition,

$$\Phi_t = \beta \Phi_{t+1} [(1 - \tau_{t+1}^k)(f_k - \delta) + 1],$$

is a constraint on the government — but written *backwards* it says

$$\Phi_{t-1} = \beta \Phi_t [(1 - \tau_t^k)(f_k - \delta) + 1].$$

To know what it is allowed to do today, the date- t

government must remember Φ_{t-1} : the **marginal value of a promise it made yesterday**.

So the minimal state is not k_t . It is (k_t, Φ_{t-1}) .

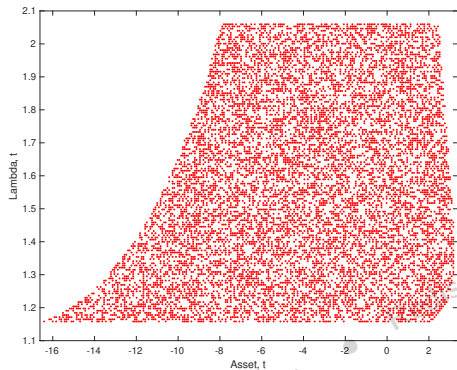
Two costs

Dimension. One more continuous state — and unlike k , its natural domain is *unknown*, which is a genuine problem when you build a grid.

Interpretation. Φ_{t-1} is not a physical object. It is the weight of past promises, carried purely to keep the government honest.

The date-0 problem is special: Φ_{-1} does not exist, so period 0 is unconstrained. That asymmetry *is* the time-inconsistency problem, written in the notation.

Research in Practice: What That State's Domain Looks Like



Each dot is an admissible state. Horizontal: the government's bond position. Vertical: the promised marginal utility — the figure's λ is this course's Φ , the Ramsey co-state. Traced by rejection sampling with a learned classifier, not by gridding.

Research in Practice

The previous slide said the co-state's domain is *unknown*. Here it is, computed — the admissible set Ω under commitment in incomplete markets.

Feng, Han, and Zhu, "Ramsey Taxation, Public Debt Sustainability, and Asset Pricing in Incomplete Markets" (under review, *Journal of Economic Theory*).

Both frontiers are economics. Promise *less* marginal utility and you promise more consumption: lower taxes, more labour, more debt households can carry — the asset limit slides left, monotonically. Promise *more* and revenue rises but the bond price falls with it, so the debt limit is **hump-shaped**: a dynamic Laffer curve.

Chamley–Judd: Zero Capital Taxation in the Long Run

The result

In a representative-agent growth model with commitment, if the Ramsey allocation converges to a steady state, the optimal **capital income tax converges to zero**.

Judd (1985, *Journal of Public Economics*); Chamley (1986, *Econometrica*).

Read the fine print

The theorem says the tax converges to zero — **if a steady state is reached**. It says nothing about how long that takes, and the optimal transition typically starts with a very high capital tax.

The intuition. A constant capital tax τ^k drives a wedge into every Euler equation between now and date t . Compounded, the implied tax on consumption at a distant horizon grows *without bound*.

An infinitely growing distortion cannot be optimal, so in the long run the wedge must vanish. Tax labour instead, and use the capital tax only in the transition.

It also assumes **full commitment**, a **representative agent**, no borrowing constraints, and a complete set of tax instruments. Every one has been shown to overturn or weaken it: commitment goes in this deck, heterogeneity and incomplete markets in **S6** and **S8**.

Without Commitment, Policy Is a *Rule*

Timing within a period, given the state K :

1. The **current** government moves first and announces $\Lambda = \{\tau^k, \tau^l, G\}$.
2. The private sector responds, having formed beliefs about
 - the actions of all **future** governments, $\tilde{\Lambda}_+ = \Lambda_+(K_+)$;
 - the resulting **law of motion** $K_+ = \tilde{\Phi}(K, \Lambda)$.
3. Households choose (c, l, a_+) optimally given those beliefs.

Because there is no commitment device, policy cannot be a *path*. It has to be a **function of the state** — a rule that the government's own future self will actually want to follow.

A game, not a program

Model the economy as a **sequential game** among a sequence of governments, none of which can bind its successors.

The solution concept is **Markov-perfect equilibrium**: strategies depend only on the payoff-relevant state.

Note what this buys: the artificial co-state Φ_{t-1} disappears. Nothing is promised, so nothing must be remembered.

Markov-Perfect Equilibrium, Recursively

Given households' best responses to any current choice $\hat{\Lambda}$ and perceived future rule $\tilde{\Lambda}$, the government's problem is

$$\hat{W}(K; \tilde{W}, \tilde{\Lambda}) = \max_{\{\tau^k, \tau^l, G\}} \left\{ U(c, G) - h(l) + \beta \tilde{W}(K_+; \tilde{\Lambda}_+) \right\}$$

subject to

$$c + K_+ - (1 - \delta)K + G \leq F(K, l),$$

$$G \leq \tau^k(r - \delta)K + \tau^l w l,$$

$$U_c = \beta [1 + (1 - \tau_+^k)(r_+ - \delta)] U_{c_+},$$

$$(1 - \tau^l) w U_c = h_l,$$

$$K_+ = \hat{\Phi}(K, \Lambda).$$

Read the third constraint

The household's **Euler equation** is a **constraint on the government** — and it contains τ_+^k , next period's tax.

So today's government is constrained by what it expects its own successor to do.

$\tilde{W}$ appears on the right-hand side but is not chosen: it is the continuation value *generated* by the equilibrium rule. Finding it is the fixed point.

Consistency: The Last Fixed Point of the Session

An MPE requires three things to agree at once:

1. **Households best-respond** to the current policy and their beliefs: $\{c, l, a_+\}(K; \hat{\Lambda}, \tilde{\Lambda}, \tilde{\Phi})$.
2. **Perceived policy is actual policy:**

$$\tilde{\Lambda}(K) = \hat{\Lambda}(K) = \Lambda(K).$$

3. **Perceived transition is actual transition:**

$$\tilde{\Phi}(K, \Lambda) = \Phi(K, \Lambda).$$

The same template, one level up

1.B1: the household's forecast of K' had to be the one its own choices produce.

Here the **government's** policy rule must be a best response to itself. A fixed point in *functions*, not in numbers.

And the object being fixed is a policy *function*, so there is no reason for it to be continuous, differentiable, or unique. That is the next slide.

The Generalized Euler Equation — and Why It Does Not Save Us

Take the cleanest case with the same structure: a decision maker with a time-consistent rule $k_+ = g(k)$, continuation value V (discount δ) and current value W (discount $\beta\delta$). Differentiating and substituting the first-order condition gives the **generalized Euler equation**

$$u'(c_0) = \beta\delta u'(c_1)f'(k_1) + (1 - \beta)\delta u'(c_1)g'(k_1).$$

- The first term is the ordinary Euler equation: $\beta = 1$ kills the second and returns it exactly.
- The second is the strategic content — today's k_1 moves what the **successor will choose**, and $g'(k_1)$ is that feedback.

Feng & Santos. The tax problem has the same shape, with $\tau'(K_+)$ in place of $g'(k_1)$ — Klein, Krusell & Ríos-Rull (2008).

Now try to find a steady state

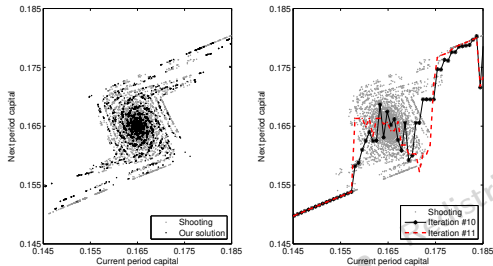
Impose $k_0 = k_1 = k$:

$$1 = \beta\delta f'(k) + (1 - \beta)\delta g'(k).$$

Solve it for $g'(k)$, substitute back through the envelope and first-order conditions, and **every term cancels**: you are left with $1 = 1$.

In the ordinary growth model $f'(k^*) = 1/\delta$ pins k^* down and shooting methods work. Here there is **nothing to linearize around**: global methods are a requirement, not a preference. That is S5 and S8.

Theory Warns Us: Santos (2002)



The Santos (2002) example computed numerically: equilibrium paths (k_t vs. k_{t+1}) do not settle on a smooth policy function.

What the example shows

In a growth model with a simple *state-dependent* capital tax rule, a **continuous** Markov equilibrium can fail to exist.

The middle steady state is unstable, and equilibrium paths cycle rather than converge to a smooth policy.

This is not a numerical artifact to be tuned away. It is a property of the model — and any method that *assumes* a smooth policy function will return a confident, wrong answer.

Research in Practice: Where This Goes Next

Research in Practice

With a representative agent, time-consistent fiscal policy over the business cycle is solvable — with global methods and care about existence.

Instructor's *Quantitative Economics* (2015); Feng & Santos, "Markov Equilibria with Quasi-Hyperbolic Discounting" (working paper), on constructive existence for these games; Klein & Ríos-Rull (2003), Klein, Krusell & Ríos-Rull (2008) for the recursive benchmark.

With heterogeneous agents the state is no longer K but the entire distribution Γ , and the government needs to know how *all* of it responds to taxes.

Research in Practice

Feng, Han, Sargent and Zhu (2026) solve that model by **deep reinforcement learning**, and characterize the distribution's evolution theoretically — a discretized Kolmogorov forward equation for $\Gamma_+ = \Phi(\Gamma)$.

The cliffhanger

2.B1 closed on it: once policy varies, aggregation needs an **extra** consistency condition — the law of motion must hold for *every* policy the government might choose. That is why the moment shortcut fails here, and why S8 uses a network.

Takeaways

- A government that can **commit** solves a Ramsey problem: maximize welfare subject to households' own optimality conditions. The price is a co-state Φ_{t-1} — the remembered weight of past promises — with an unknown domain.
- **Chamley–Judd** says the long-run capital tax is zero, *if* a steady state is reached, *if* agents are identical, *if* markets are complete. Relax any one and the result weakens.
- Without commitment, policy is a **rule** $\tau(K)$, and equilibrium is a fixed point in functions: the rule must be a best response to itself.
- The **generalized Euler equation** involves the derivative of that unknown rule, so steady states cannot be read off it — local methods lose their anchor and global methods become mandatory.
- **Santos (2002)**: a continuous Markov equilibrium may not exist at all. Assume smoothness and your solver will answer anyway.

Bridge: Session 2 in One Slide — and Next Week

Model	Aggregate state	Consistency condition
Huggett	λ , constant	bond holdings net to zero
Aiyagari	λ , constant	λ invariant; $K = \int a d\lambda$
Krusell–Smith	Γ and Z	$\Gamma' = H(\Gamma, Z, Z')$ is the one choices generate
Life-cycle OLG	cohorts $\Gamma_1, \dots, \Gamma_I$	cohort aggregates clear markets
Taxation without commitment	K (with heterogeneity, Γ)	$\tau(\cdot)$ is a best response to itself

Not one of these can be solved with a pencil. Every row is a fixed point in an infinite-dimensional space.

Next week (S3)

What it means to *solve* a model: discretize, approximate, iterate, check — plus the Python to run it.

Before then: Ljungqvist–Sargent ch. 17–18 or Heer–Maussner ch. 7; install Python with the setup lab.